

FRANK PORTER
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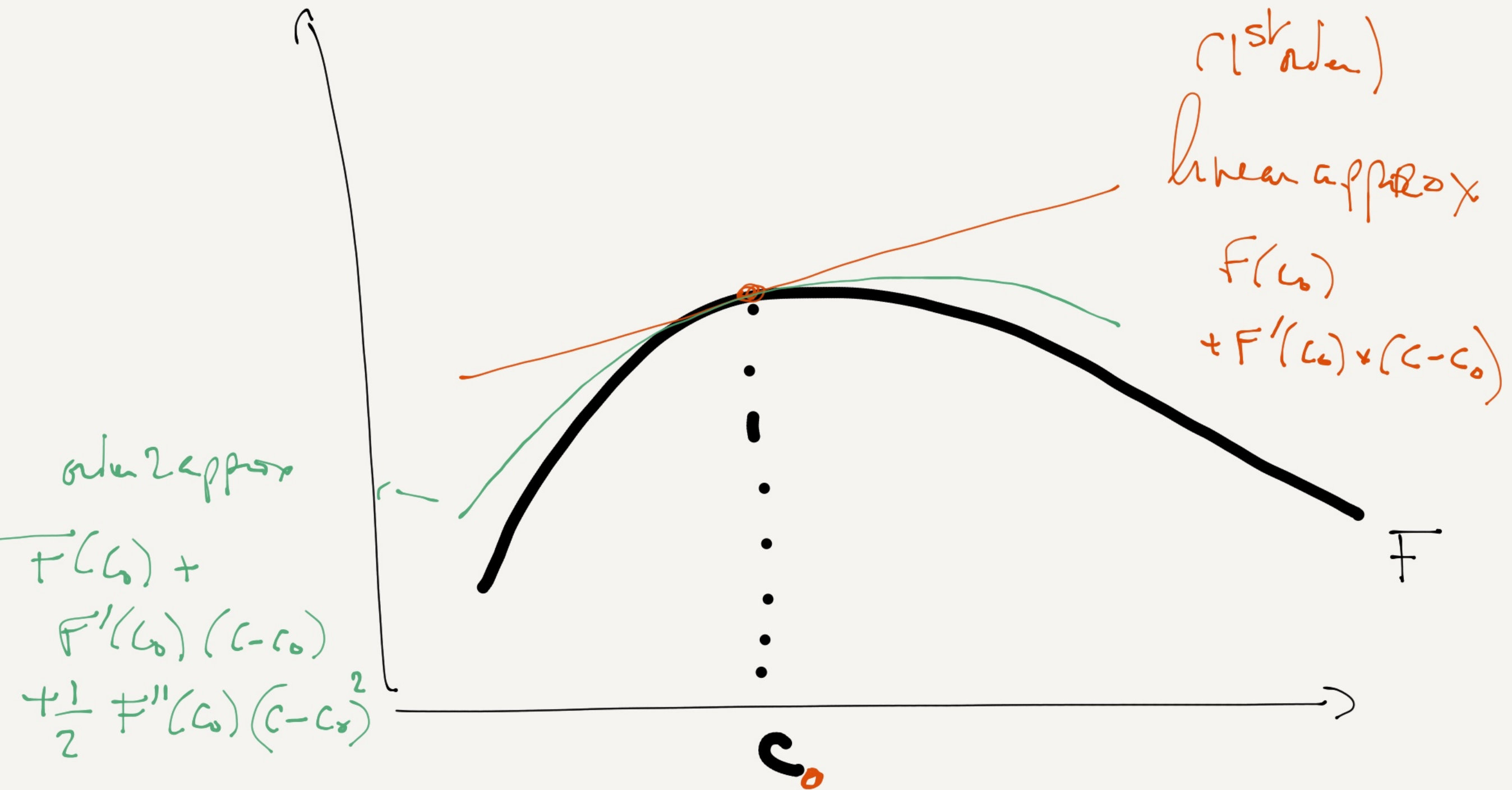
M1 MACRO

Group B

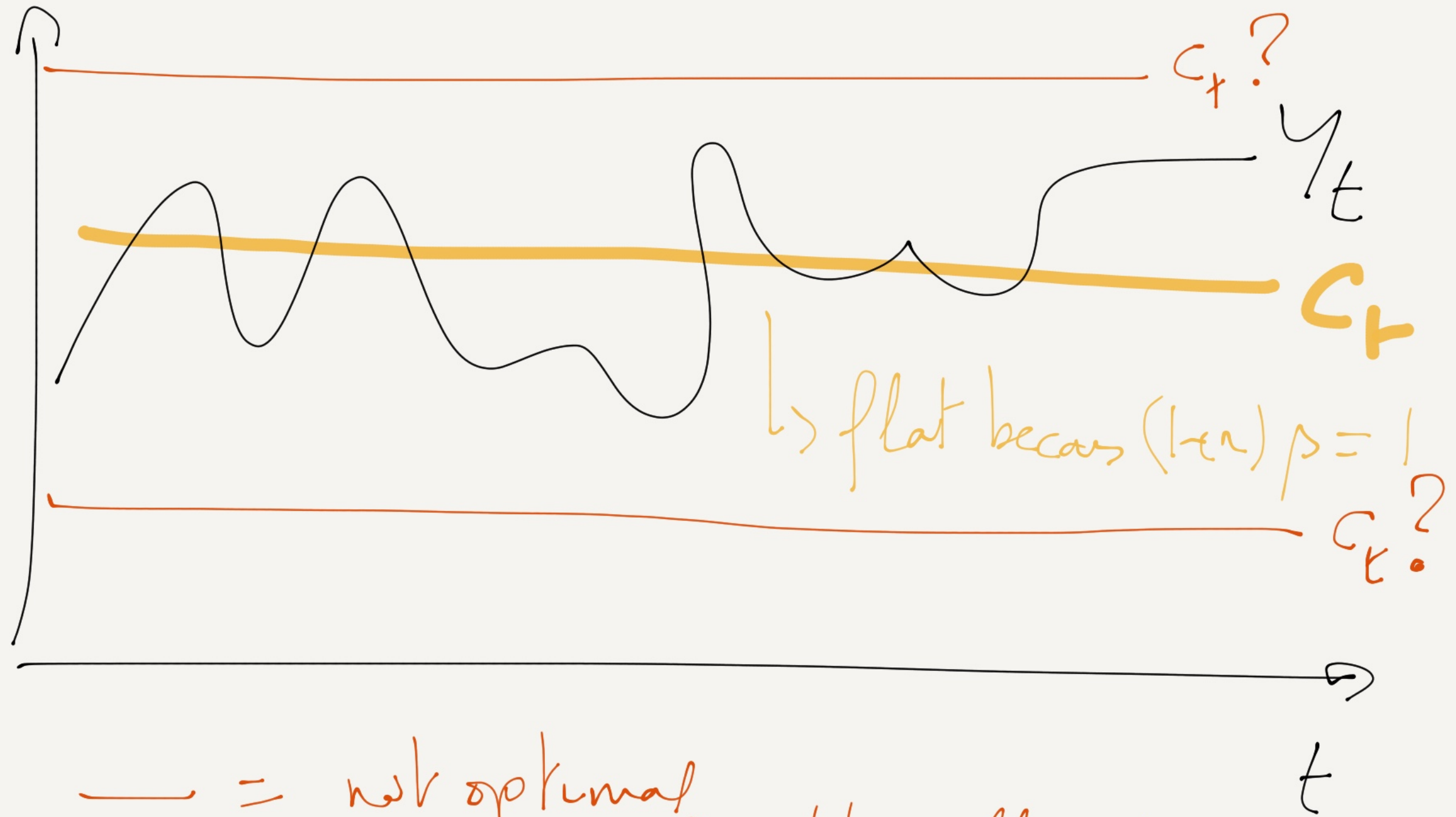
Session 8

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

TAYLOR EXPANSIONS



Deterministic model



— = not optimal
 — optimal or not ~~feasible~~ allowed

$$A_{t+1} = (1+r)A_t - C_t + Y_t$$

$$\Rightarrow A_t = \frac{1}{1+r} A_{t+1} - \frac{Y_t}{1+r} + \frac{C_t}{1+r}$$

↳ also true is $t+1$: $A_{t+1} = \frac{1}{1+r} A_{t+2} - \frac{Y_{t+1}}{1+r}$

$\Rightarrow A_t = \frac{1}{1+r} \left[\frac{1}{1+r} A_{t+2} - \frac{Y_{t+1}}{1+r} + \frac{C_{t+1}}{1+r} - \frac{Y_t}{1+r} + C_t \right]$

↳ $A_{t+2} = \frac{1}{1+r} A_{t+3} - \frac{Y_{t+2}}{1+r} + \frac{C_{t+2}}{1+r}$

$$\Rightarrow A_t = \left(\frac{1}{1+r}\right)^T A_{t+T} + \frac{1}{1+r} \sum_{j=0}^{T-1} \left(\frac{1}{1+r}\right)^j (C_{t+j} - Y_{t+j})$$

$$A_t = \lim_{T \rightarrow \infty} \left(\frac{1}{1+r} \right)^T A_{t+T} + \lim_{T \rightarrow \infty} \sum_{j=0}^{T-1} \left(\frac{1}{1+r} \right)^{T-1-j} (c_{t+j} - y_{t+j})$$

- * not optimal to be > 0
- * not allowed to be < 0

Hence $\lim_{T \rightarrow \infty} \left(\frac{1}{1+r} \right)^T A_{t+T} = 0$

= TRANSVERSALITY CONDITION
(TC)

(Intuition:
Assume that the world ends at $t+T-1$)

$$A_t + H_t = \frac{1}{1+r} \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j C_{t+j}$$

$$C_t + A_{t+1} \stackrel{!}{=} (1+r)A_t + Y_t$$

From the Euler eq., $C_{t+j} = C \quad \forall j$

$$\rightarrow A_t + H_t = \frac{C}{1+r} \times \underbrace{\sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j}_{= \frac{1}{1 - \frac{1}{1+r}}} \quad (2.28)$$

$$A_t + H_t = \frac{C}{r} \Rightarrow \boxed{C = r(A_t + H_t)}$$